



The association between depressive symptoms and executive function in type 1 diabetes population: A scoping review

Chawisa Dandamrongrak^{a,b,*}, Kaitlyn Rechenberg^a, Jumpee Granger^b, Ayesha Johnson^a, Kailei Yan^a, Jennifer Kue^a

^a College of Nursing, University of South Florida, USA

^b Ramathibodi School of Nursing, Faculty of Medicine, Ramathibodi Hospital, Mahidol University, Thailand

ARTICLE INFO

Keywords:

Executive function
Depression
Depressive symptoms
Type 1 diabetes mellitus

ABSTRACT

Purpose: The purpose of this study was to explore the existing literature on the relationship between depressive symptoms and executive function in patients with type 1 diabetes (T1DM) across the lifespan.

Methods: The scoping review followed the PRISMA protocol by using three databases: PubMed, CINAHL, and PsycINFO on May 14, 2023. Primary research that included reported executive function and the association with depressive symptoms was included in the review.

Results: Of 1470 de-duplicated publications identified, nine articles were included in the review. Five studies evaluated the T1DM population, while four studies evaluated T1DM and type 2 diabetes (T2DM) as an aggregate result. Three studies indicated an association between depressive symptoms and executive function in adults with T1DM, and four studies indicated an association between depressive symptoms and executive function in adults with either T1DM or T2DM. In general, participants who reported depressive symptoms also exhibited poor executive function. However, two studies did not find an association between depressive symptoms and executive function.

Conclusion: In summary, the seven studies in this review suggest that individuals with T1DM who report depressive symptoms are at a higher risk of poor executive function; a clear association between depressive symptoms and executive function in individuals with T1DM remains inconclusive. There is a need to explore this relationship in the future.

1. Introduction

More than eight million people are living with type 1 diabetes mellitus (T1DM) globally, with an estimated 520,000 new diagnoses reported in 2022 [1]. There were an estimated 182,000 deaths due to T1DM in the same year, highlighting the ongoing challenge of managing and addressing this chronic medical condition [1].

It is well-known that patients with T1DM have a risk of depressive symptoms across their lifespan [2–6]. Depressive symptoms are prevalent among adolescents with T1DM, estimated to be 2–3 times higher than among adolescents without T1DM [7], and 22.3% of adults with T1DM (out of a sample size of N=325) have clinically significant depressive symptoms [2]. These symptoms have a bidirectional relationship with diabetes self-management, leading to poorer diabetes outcomes across the lifespan [8–11]. Depressive symptoms are associated with a high level of diabetes family conflict, high diabetes burden,

low quality of life [12], poor diabetes self-management, hyperglycemia [13–15], and poor sleep quality [16,17]. Additionally, the presence of depressive symptoms increases the risk of developing neurocognitive deficiencies in memory, learning, and executive function, which are important in diabetes care [11].

Executive function refers to mental processes involving concentration, attention, working memory, and cognitive flexibility [18]. In addition, executive function is a set of mental skills that are used to develop reasoning, problem-solving, and planning [18]. The core components of executive function are inhibition (including self-control and interference control), working memory, and cognitive flexibility (also known as set-shifting, or mental flexibility) [18]. Higher-order executive functions such as reasoning, problem-solving, and planning are constructed from these skills [18]. Therefore, executive function is one of the essential factors in succeeding in T1DM self-management.

However, individuals with diabetes face a 1.5–2.5 times higher risk

* Correspondence to: 12912 USF Health drive, Tampa, FL, 33612, USA.

E-mail address: Chawisa@usf.edu (C. Dandamrongrak).

<https://doi.org/10.1016/j.pcd.2024.04.001>

Received 18 October 2023; Received in revised form 13 February 2024; Accepted 8 April 2024

Available online 22 April 2024

1751-9918/© 2024 Primary Care Diabetes Europe. Published by Elsevier Ltd. All rights reserved.

of cognitive decline compared to the healthy population [19]. Poor executive function is prevalent across the lifespan of those with T1DM [20,21]. The literature shows that individuals with T1DM have poorer overall cognition and poorer executive function compared to their peers without T1DM [20–23]. It is important to note, however, that this relationship is not merely causal; rather, it is complex and bidirectional [24]. This is in part because many diabetes-related factors can lead to poor executive function. For example, hyperglycemia is the main factor that impairs executive function in individuals of diverse age groups and ethnic backgrounds in both T1DM and type 2 diabetes (T2DM) [24]. At the same time, individuals with poor executive function have difficulties in diabetes self-management, resulting in persistent hyperglycemia and a low quality of life [25,26]. Within the relationship between T1DM and executive function, depressive symptoms can exacerbate this interplay, thereby compounding the negative impact on poorer executive function [24]. Similar to other populations, such as healthy adults, cancer patients, and patients with depression, depressive symptoms are associated with poor cognitive function across multiple domains, especially poor executive function [27–31].

An understanding of this complex relationship may be important to

future research in T1DM. In a study with cancer patients, the association between depressive symptoms and executive function suggests the necessity of incorporating routine neuropsychological assessments alongside depressive symptom measurements [31]. This approach not only refines prognosis but also facilitates the development of psychological and cognitive interventions [31]. A similar understanding of how depressive symptoms impact executive function among individuals with T1DM may also be beneficial. The purpose of this scoping review is to explore the existing literature on the relationship between depressive symptoms and executive function in individuals with T1DM across the lifespan. To guide the exploration and mapping of the existing literature, the research question was developed based on the PECO (Population, Exposure, Comparison, Outcome) framework [32]: In individuals with T1DM, does the presence of depressive symptoms impact executive function across the lifespan? This understanding may improve diabetes self-management and contribute valuable scientific knowledge to the studies of T1DM.

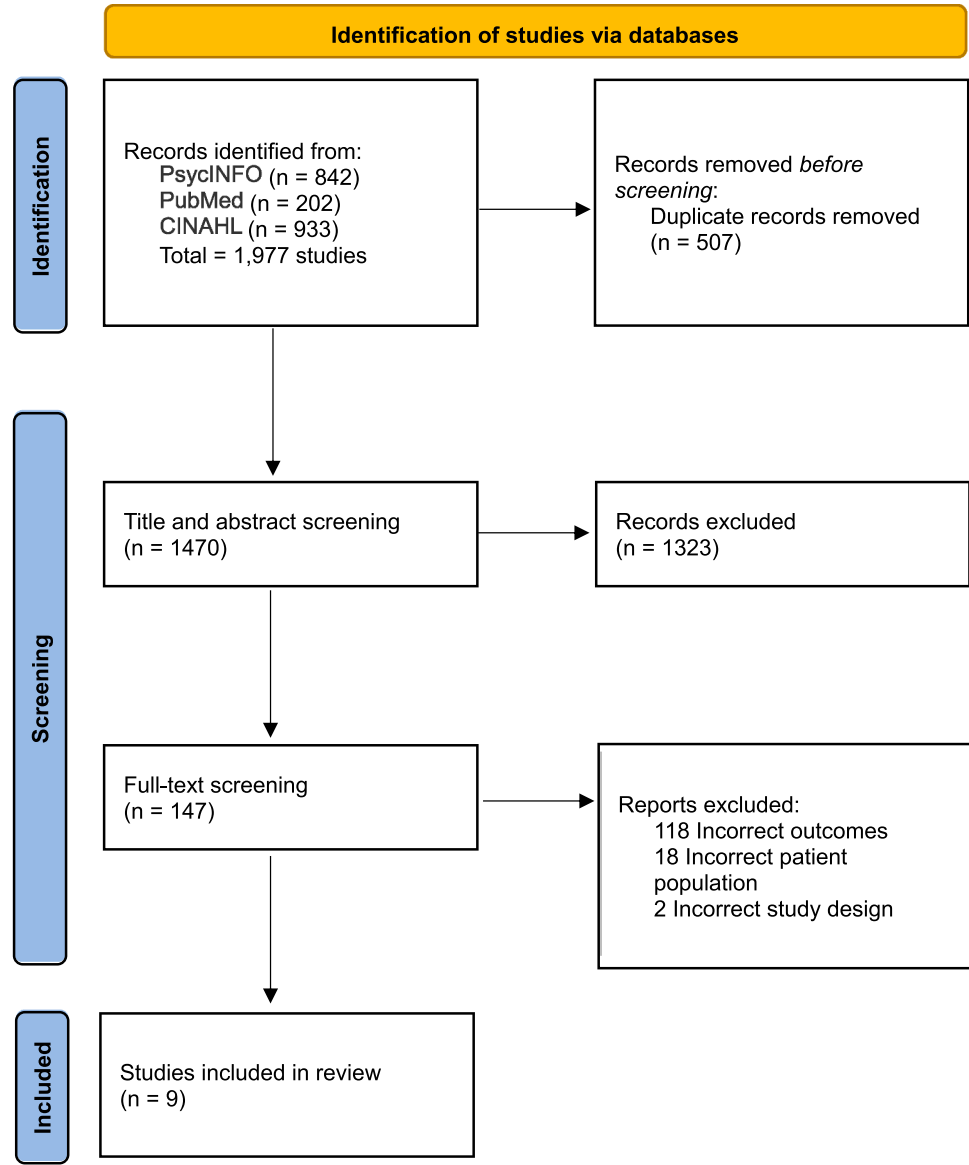


Fig. 1. PRISMA figure. PRISMA figure. J. Page., et al., 2021, *BMJ*, 372, 71. Copyright [2023] by the PRISMA.
(a) Adapted from “The PRISMA 2020 statement: An updated guideline for reporting systematic reviews,” by M. (b) Adapted with permission.

2. Methods

This review was conducted based on the methodology described in the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews) [33] as shown in Fig. 1.

2.1. Inclusion criteria

The eligible studies included primary peer-reviewed studies written in English. The target study population was people with T1DM throughout their lifespan. All articles that included depressive symptoms or executive function in the title or abstract text were eligible for review. All published articles that met the eligibility criteria were reviewed. Gray literature, theses, dissertations, and literature reviews, such as scoping, systematic, and integrative reviews, were excluded.

2.2. Search strategy

Our systematic review of the literature was conducted from May 4, 2023, to July 24, 2023. The search was conducted using the Medical Literature Analysis and Retrieval System Online (MEDLINE) via the free interface PubMed, Cumulative Index to Nursing and Allied Health (CINAHL), and PsycINFO on May 14, 2023. The search terms were identified with the help of a health sciences librarian. The combination of the major search terms, including *depression*, *depressive symptoms*, *executive function*, and *type 1 diabetes*, was used similarly across the three databases. However, the keywords were expanded in each database to maximize the number of articles by adding the associated MeSH terms and associated subject headings.

2.3. Selection process

All articles that met the eligibility criteria were uploaded to Covidence systematic review software (available at www.covidence.org). Duplicates were identified and automatically removed before the screening process began. The screening process included (1) title and abstract screening and (2) full-text screening. The studies were reviewed based on the inclusion criteria listed above. Two authors (CD and KY) worked independently to screen all articles, and a third author (JK) assisted in resolving any conflicts that arose.

2.4. Data extraction and synthesis

A preliminary data extraction table was created in Microsoft Word. The extracted data included the authors' names and publication year, study design, study objectives, sample description, instruments, results, and study limitations. The form was modified as needed throughout the extraction process. The data extraction table was evaluated collaboratively by two authors (CD and KY), who discussed the relevance to the research questions of the scoping review and resolved conflicts as needed.

3. Results

The literature search yielded 1977 publications. After removing 507 duplicates, 1470 articles remained for a title and abstract screening. Following a full-text screening of 147 articles, 37 were selected for extraction. Of those, 21 studies were excluded due to the absence of evidence of a relationship between depressive symptoms and executive function. Two articles were removed based on the incorrect target population, and five studies were eliminated for not evaluating executive function. The remaining nine studies included in the analysis for this review are described in Table 1.

3.1. Study characteristics

Of the nine studies included in the analysis, five studies used a cross-sectional design [34–38], while four studies used a longitudinal design [39–42]. The age range of the entire sample was 13–89 years old. Five studies specifically focused on individuals with T1DM [34–37,40], while four studies did not identify type of diabetes in samples [38,39,41,42]. However, since T1DM patients were also examined in the general diabetes samples, these four studies were included in the review. The total sample size across all nine studies was 15,557 participants, including 438 individuals with T1DM (2.8%), 11,377 individuals with T1DM or T2DM (73.13%), and 3492 healthy participants in the control groups (22.45%).

Measures of depressive symptoms and executive function were not consistent across the studies as shown in Table 2. The studies used various self-administered instruments to measure depressive symptoms. Executive function was measured using different types of instruments including parent-reports, self-reports, and performance-based instruments.

3.2. Depressive symptoms and executive function

Overall, seven articles [34,35,38–42] from the reviewed studies reported a relationship between depressive symptoms and poor executive function in individuals with T1DM and two studies did not [36,37].

In bivariate analysis, more depressive symptoms were associated with poor executive function in adolescents aged 18 years old, young adults, and adults with T1DM ($p < .01$) [34,35]. Four studies focusing on both T1DM and T2DM adults found that having depressive symptoms was associated with poorer general cognitive function, specifically in the domains of attention and memory, which are components of inhibition and working memory in executive function ($p < .05$) [38,39,41,42].

The inclusion of depressive symptoms and other psychosocial factors (i.e., eating disorders, stress, anxiety, social support, and exercise behavior) in the regression model resulted in a decrease in the statistical significance ($p < .001$) of the association with working memory domain and general cognitive function to non-statistical significance ($p = .92$) [34,40]. However, it was observed that more depressive symptoms could improve the model fit by 6.7%, indicating that depressive symptoms can predict poorer executive function, which is a component of general cognitive function [40]. These results indicated the adults with T1DM who experienced depressive symptoms also had poor executive function.

One study focusing on adolescents (aged 13–17 years) found no association between depressive symptoms and executive function in adolescents with T1DM in bivariate analysis [37]. Another study in adults demonstrated no direct relationship between depressive symptoms and executive function or working memory when examined individually [36]. However, in the same study, adults who self-reported experiencing depressive symptoms exhibited poorer general cognitive ability ($\beta = -0.194$, $p < 0.05$) [36].

4. Discussion

Overall, the results regarding the association between depressive symptoms and poor executive function in individuals with T1DM were inconclusive. Seven studies indicated an association between depressive symptoms and poorer executive function across different age groups. Two studies did not find an association.

The studies focusing on older adolescents and adults (aged ≥ 18 years old) found an association between more depressive symptoms and poorer executive function [34,35,40]. Additionally, adults with either T1DM or T2DM also exhibit poor overall cognitive function [36,38,39,42]. These results align with broader research indicating that the development and expression of depressive symptoms may contribute to impaired executive function in various populations [43–45]. The

Table 1
Variables of Interest of Nine Studies.

Study (authors, date)	Study Design	Study objective	Sample Description	Instruments	Results	Limitations of the study
Broadley et al. (2018)	A cross-sectional study	To determine the association between executive function and disordered eating in young adults with type 1 diabetes (T1DM), relative to a control group without diabetes	N = 74 participants aged between 18 and 40 years, diagnosed with T1DM for at least 12 months. N= 201 demographically similar control participants	1. Behavior Rating Inventory of Executive Function-Adult version (BRIEF-A) 2. Depression Anxiety and Stress Scales (DASS)	Bivariate correlations between depression anxiety and stress and executive function revealed that the DASS total score and the BRIEF total scores were significantly positively correlated ($p < .01$).	The control group sample's demographics included a substantial percentage of current university students (79.6%). Given that there are more mental disorders in this population than in the general population, comparing with this population may pose a risk of bias. Additionally, using performance-based tests may result in a more accurate measurement of executive function.
Perez et al. (2017)	A cross-sectional study	To examine parent reports of executive function to measures of adherence, glycemic control (A1C), and psychosocial outcomes (depression and quality of life) in adolescents with T1DM.	N = 120 adolescents aged between 13 and 17 years, diagnosed with T1DM for at least 6 months, no other major health problems, not participating in other intervention studies, and A1C level between 8.0 and 12.0% at the time of recruitment.	1. Parents completed the Behavior Rating Inventory of Executive Function 2. The Patient Health Questionnaire (PHQ-9).	No significant association between executive function and depressive symptoms.	Data from a questionnaire provided by parents were used to assess executive function. While the BRIEF is a well-known tool for this purpose, it relies on parent reports. It might be more effective to utilize data reported directly by adolescents themselves.
Bell, et al. 2017	A longitudinal population-based study	To investigate the association between cognitive decline and incident depressive symptoms in older Puerto Rican adults with diabetes over four years.	N = 680 participants aged 60–89 years old with diabetes	1. The Mini-Mental Cabán (MMC) 2. Geriatric Depression Scale (GDS)	In a covariate-adjusted logistic regression model, cognitive decline, female gender, and greater diabetes-related complications were each significantly associated with increased odds of incident depressive symptoms ($p < .05$). In a multiple regression model adjusted for covariates, incident depressive symptoms and older age were associated with greater cognitive decline, and higher education was related to less cognitive decline ($p < .05$).	This study utilized only two time points for assessments. Therefore, potential non-linear trends could not be examined over time. Additionally, participants who had cognitive impairment at baseline were excluded. As a result, the relationships between depressed symptoms and cognitive functioning may be underestimated. In this study, no differentiation was made between type 1 and type 2 diabetes. Since the etiologies of type 1 and type 2 diabetes are different, as well as the recommended self-management or treatment regimens, factors influencing the link between diabetes and cognitive function may vary based on the type of diabetes.
Bikri et al., 2021	A cross-sectional study	To determine whether socio-demographic and clinical characteristics, eating attitude, and psycho-emotional characteristics can predict visual perception and working memory impairment in the Moroccan population with T1DM.	N=140 adults aged 46.75±8.92 years old with T1DM.	1. The digital version of the Rey-Osterrieth Complex Figure Type A test (ROCF-A). 2. The 21-item Depression, Anxiety, and Stress Scale (DASS-21).	A highly significant negative correlation between mental health characteristics (stress, anxiety, and depression) and visual perceptions ($p < .001$) and working memory ($p < .001$) was detected in Pearson's correlation test. However, mental health characteristics (stress, anxiety, and depression) were not statistically significant for both	This study evaluated working memory, which is only one component of executive functions. However, it may not fully represent the overall executive function in this population. Additionally, there was no statistically significant difference in the regression model, which could be due to multicollinearity among predictors in the model.

(continued on next page)

Table 1 (continued)

Study (authors, date)	Study Design	Study objective	Sample Description	Instruments	Results	Limitations of the study
Chiu et al., 2016	The population-based longitudinal study.	To examine levels and rates of change in age trajectories of cognitive function in middle-aged and older adults with and without diabetes who participated in different health behaviors.	N = 4076 participants aged 53 and above with and without diabetes from the 1999 Taiwan longitudinal study of aging. Including N = 584 participants with diabetes, and 3492 without diabetes. The mean age of the sample was 69.3 years (SD = 9.1).	1. The 9-item Short Portable Mental Status Questionnaire (SPMSQ) 2. The Center for Epidemiologic Study Depression Scale (CESD)	visual perceptions and working memory when added to the regression model. Adjusting for exercise, social support and depressive symptoms reduce the association between diabetes and levels of cognitive function to non-significance, and each factor improved the model fit by as large as 33.0%, 7.0%, and 6.7% respectively. These results indicate a complete mediating effect of exercise, support, or depressive symptoms on the association between diabetes and levels of cognitive function.	The attrition of our baseline sample was 18.4%. Individuals who passed away or were lost to follow-up were older, less educated, and had lower cognitive performance, leisure activities, exercise, support, and poorer health status. Additionally, they exhibited higher depressive symptoms at baseline, which were variables related to cognitive functioning in the study. Moreover, they had lower levels of leisure activities, higher stress, less smoking, and different diet behaviors. As a result, this study may underestimate the relationship between modifiable variables and cognitive function. In this study, no differentiation was made between type 1 and type 2 diabetes. Since the etiologies of type 1 and type 2 diabetes differ, as well as the recommended self-management or treatment regimens, factors influencing the link between diabetes and cognitive function may vary depending on the type of diabetes.
Downer et al., 2019	Retrospective cohort study with longitudinal analysis.	To assess the relationship between comorbid depression and diabetes and cognitive decline among Mexican Americans aged 65 and over.	N = 2756 Americans age 73.3±6.5 with diabetes and over-residing in Texas, New Mexico, Colorado, Arizona, and California.	1. The Mini-Mental State Examination (MMSE). 2. The Epidemiologic Studies Depression Scale (CES-D)	Participants with depression only or comorbid depression and diabetes had significantly lower baseline cognitive functioning compared to participants with no depression and no diabetes (P <0.05). In general, participants with comorbid depression and diabetes had lower scores on the MMSE at each examination wave compared to the other comorbidity categories and these differences appeared to increase over time. Throughout the six observation waves, participants with comorbid depression and diabetes declined 6.5 points on the MMSE, whereas participants with depression declined by 4.4 points, and	This study used self-reporting to indicate the diagnosis of diabetes. Additionally, they utilized a cut point of 16 on the CES-D scale to classify individuals as depressed or not depressed. This approach may result in incorrect classification. The MMSE measures only one domain of executive function, which is attention. Therefore, the results cannot fully represent the executive function in this sample. In this study, no differentiation was made between type 1 and type 2 diabetes. Since the etiologies of type 1 and type 2 diabetes differ, as well as the recommended self-management or treatment regimens, factors influencing the link between diabetes and cognitive function may vary depending on the type of diabetes.

(continued on next page)

Table 1 (continued)

Study (authors, date)	Study Design	Study objective	Sample Description	Instruments	Results	Limitations of the study
Duinkerken et al., 2016	A cross-sectional study	To assess the effect of T1DM and concomitant proliferative retinopathy, as a marker of cumulative hyperglycemic exposure, on cognitive functioning, brain structure, and functional and structural connectivity.	N = 104 T1D patients and N = 49 controls without diabetes and no psychiatric diagnosis or treatment. Participants aged between 18 and 56 years old.	1. The Epidemiologic Studies Depression Scale (CES-D) 2. The total time to complete part 3 of the Stroop Color-Word Test corrected for the mean time of parts 1 and 2 3. The total time of Concept Shifting Task (CST) part C corrected for the mean time of parts A and B 4. The total amount of wrongly checked letters on the D2 test 5. The total amount of perseverative errors made on the Wisconsin Card Sorting Test (WCST) 6. The total amount of animals named in 2 minutes and insects in 1 minute on the Category Word Fluency Task	participants with diabetes declined by 7.8 points, and participants without depression or diabetes declined by 4.2 points. In all participants, after adjustment for age, sex, and systolic blood pressure, more self-reported depressive symptoms correlated with lower general cognitive ability scores ($\beta = -0.194$, $p < 0.05$). Cognitive domains, such as memory and executive functions, were not related to depressive symptoms.	There was no clinical diagnostic used to confirm depression in this study. Additionally, the study did not include individuals receiving treatment for depression or any other psychiatric disease. Therefore, depression may be underrepresented in this sample.
Rotkiewicz-Piorun et al., 2006	A Longitudinal study	To examine predictors of cognitive decline over seven years among older Mexican Americans with diabetes.	N = 808 The Hispanic Established Population for the Epidemiological Study of the Elderly (HEPESE), a longitudinal study of Mexican Americans aged ≥ 65 , residing in Texas, New Mexico, Colorado, Arizona, and California. The mean age of the sample was 72.5 ± 5.8 (SD).	1. The Mini-Mental State Examination (MMSE). 2. The Center for Epidemiologic Studies Depression Scale (CES-D).	High depressive symptoms (estimate = -0.03, SE = 0.01) was significantly associated with lower MMSE scores at each follow-up. Which means more depressive symptoms were correlated with poorer cognitive ability.	The MMSE measures only one domain of executive function, which is attention. Therefore, the results cannot fully represent the executive function in this sample. In this study, no differentiation was made between type 1 and type 2 diabetes. Since the etiologies of type 1 and type 2 diabetes differ, as well as the recommended self-management or treatment regimens, factors influencing the link between diabetes and cognitive function may vary depending on the type of diabetes.
Yin et al., 2023	A cross-sectional study.	To investigate the effects of depression on cognitive function in patients of different ages with diabetes mellitus (DM).	N = 6549 Patients aged 49–71 years old with diabetes in the 2016 Kailuan Group staff physical examination.	1. The Mini-Mental State Examination (MMSE). 2. Self-Rating Depression Scale (SDS).	Generalized linear regression analysis showed that a higher SDS index score was associated with a lower MMSE score ($\beta = -0.06$; $P < 0.01$). Which means more depressive symptoms were correlated with poorer cognitive ability.	The MMSE measures only one domain of executive function, which is attention. As a result, the results cannot fully represent the executive function in this sample. Furthermore, this study did not differentiate between type 1 and type 2 diabetes. Given that the etiologies of type 1 and type 2 diabetes are different, along with the recommended self-management or treatment regimens, factors influencing the link between diabetes and

(continued on next page)

Table 1 (continued)

Study (authors, date)	Study Design	Study objective	Sample Description	Instruments	Results	Limitations of the study
						cognitive function may vary by type of diabetes

Table 2
The Measurement of Depressive Symptoms and Executive Function.

Measures of Depressive Symptoms	n
•Depression Anxiety and Stress Scales (DASS)	2
•Patient Health Questionnaire (PHQ-9)	1
•Geriatric Depression Scale (GDS)	1
•Center for Epidemiologic Studies Depression Scale (CES-D)	4
•Self-Rating Depression Scale (SDS)	1
Measures of Executive Function	n
•Behavior Rating Inventory of Executive Function-Adult version (BRIEF-A)	1
•Behavior Rating Inventory of Executive Function (BRIEF-Parents report)	1
•Mini-Mental Cabán (MMC)	1
•Mini-Mental State Examination (MMSE)	3
•Rey-Osterrieth Complex Figure Type A test (ROCF-A)	1
•Short Portable Mental Status Questionnaire (SPMSQ)	1
•Stroop Color-Word Test	1
•Concept Shifting Task (CST)	1
•D2-test	1
•Wisconsin Card Sorting Test (WCST)	1
•Category Word Fluency Task	1

Note. Duinkerken et al. (2016) used five instruments to measure executive function.

findings from previous studies involving children without T1DM aged 7–15 years old [43], the general population across the lifespan [44], and depression patients in symptomatic and recovery phases [45], consistently support the notion that more depressive symptoms are associated with poorer executive function. However, two studies found no association between depressive symptoms and executive function in adolescents aged 13–17 years and adults [36,37]. The conflicting findings in these reviewed studies highlight the potential methodological limitations of these studies.

Firstly, in a study on adolescents [37], only 120 adolescents aged 13–17 years were included in the study, which may not fully represent the population of adolescents with T1DM. Secondly, they conducted their study with adolescents using parent-reported measures for executive function [37], while the other studies used self-administered instruments in adults, which may result in different outcomes. Lastly, while Duinkerken and colleague did not find a direct association between depressive symptoms and executive function, they did find an association between depressive symptoms and general cognitive function [36]. This could be attributed to the use of different instruments in the study compared to those used in other included studies and the exclusion of participants who received treatment for depressive symptoms.

Despite the conflicting results, the review highlights a potential association between depressive symptoms and poorer executive function in people with T1DM. Specifically, patients with T1DM who experience depressive symptoms may be at risk of poorer executive function. However, several potential limitations in the review process should be acknowledged. Firstly, there was a wide age range across all samples, and different age ranges were covered in each of the nine studies, making it difficult to summarize the results for specific age groups. Additionally, only nine studies were included in the review, and four of them did not specify the types of diabetes. This is significant because differences in self-management of T1DM and T2DM may impact the severity of depressive symptoms and executive function. Lastly, most of the included studies measured only working memory and attention,

which may not fully capture overall executive function. Thus, these results may not represent the overall picture of depressive symptoms and executive function in the general T1DM population. These limitations emphasize the need for more comprehensive and specific investigations in this area.

Future research should continue exploring the association between depressive symptoms and executive function by using the standardized measurement of executive function and controlling for potential confounders such as age, gender, diabetes type, diabetes duration, self-management, and social support in statistical analyses. For better understanding of the impact of diabetes on the association between depressive symptoms and executive function, future research may consider expanding the study population to include both types of diabetes and comparing it with patients who do not have diabetes. This approach may contribute to a more comprehensive analysis of the relationships between these variables, providing valuable insights into the interplay of diabetes, depressive symptoms, and executive function. Moreover, it is critical to focus on specific age groups, including children and adolescents. Understanding the link between depressive symptoms and executive function in children and adolescents with T1DM can contribute to efforts aimed at preventing the decline in executive function during adulthood. Additionally, the association between depressive symptoms and executive function highlights the importance of screening for depressive symptoms within this population. This result also suggests the need for the development of interventions to accommodate patients with different learning and diabetes self-management abilities. Therefore, interventions aimed at reducing or managing depressive symptoms can be enhanced to be more effective in patients experiencing difficulties with attention, memory, or cognitive flexibility. These interventions can be tailored to address specific cognitive challenges, such as poor attention, memory deficits, or difficulty in adjustment, faced by individuals with T1DM and depressive symptoms, ultimately leading to improved diabetes self-management and quality of life.

5. Conclusion

An understanding of the association between depressive symptoms and executive function supports the importance of screening for depressive symptoms. This is important for the development of interventions and the planning of individual diabetes self-management. While seven studies have suggested that older individuals with T1DM who experience depressive symptoms are at a higher risk of poorer executive function, two studies did not find this association. These contradictions may result from methodological limitations, including small sample sizes and variations in instruments used. Therefore, a clear association between depressive symptoms and poor executive function in individuals with T1DM remains inconclusive. Researchers should consider using self-administered or performance-based instruments to measure executive function, and standardized measurements for depressive symptoms when expanding the study population. This may enhance the reliability and validity of the research findings. Additionally, analyzing executive function in specific domains, including memory, attention, and cognitive flexibility, may help us better understand individuals' development and their ability to self-manage T1DM.

Funding acknowledgment

This research did not receive any specific grant from funding

Table 3
Search strategy and combinations.

Search Term	Database		
	PEBMED	CINAHL	PSYCLNFO
Depressive Symptoms	("Depression"[Mesh] OR "Emotional Depression"[Text Word] OR "Depressive Symptom*" [Text Word]) OR "Depressive Disorder"[Mesh] OR "Depressive Disorders"[Text Word] OR "Depressive Neuroses"[Text Word] OR "Depressive Neurosis"[Text Word] OR "Depressive Syndrome"[Text Word] OR "Depressive Syndromes"[Text Word] OR "Melancholia"[Text Word] OR "Melancholias"[Text Word] OR "depressive episode*" [Text Word] OR "affective disorder*" OR "mood disorder*" OR "dysthymic disorder*" [Text Word] OR "Mood Disorders"[Mesh] OR "Seasonal Affective Disorder"[Mesh] OR "Bipolar Disorder"[Mesh] OR "Premenstrual Dysphoric Disorder"[Mesh] OR "Dysthymic Disorder"[Mesh] OR "dysthymia*" [Text Word])	(MH "Depression" OR MM "Premenstrual Dysphoric Disorder" OR (MM "Seasonal Affective Disorder" OR MH "Bipolar Disorder" OR MM "Dysthymic Disorder") OR TI (depression or depressive disorder or depressive symptoms OR "affective disorder*" OR "depressive Disorder*" OR "depressive episode*" OR "depressive neuroses" OR "depressive neurosis" OR "depressive symptom*" OR "depressive syndrome" OR "depressive syndromes" OR "depression" OR "Emotional Depression" OR "Melancholia" OR "Melancholias" OR "mood disorder*" OR "dysthymic disorder*" OR "dysthymia*") OR AB (depression or depressive disorder or depressive symptoms OR "affective disorder*" "depressive Disorder*" OR "depressive episode*" OR "depressive neuroses" OR "depressive neurosis" OR "depressive symptom*" OR "depressive syndrome" OR "depressive syndromes" OR "depression" OR "Emotional Depression" OR "Melancholia" OR "Melancholias" OR "mood disorder*" OR "dysthymic disorder*" OR "dysthymia*") OR (SU depression or depressive disorder or depressive symptoms OR "affective disorder*" "depressive Disorder*" OR "depressive episode*" OR "depressive neuroses" OR "depressive neurosis" OR "depressive symptom*" OR "depressive syndrome" OR "depressive syndromes" OR "depression" OR "Emotional Depression" OR "Melancholia" OR "Melancholias" OR "mood disorder*" OR "dysthymic disorder*" OR "dysthymia*")	(MM "Depression (Emotion)" OR MM "Major Depression" OR DE "Affective Disorders" OR DE "Disruptive Mood Dysregulation Disorder" OR DE "Major Depression" OR DE "Persistent Depressive Disorder" OR DE "Premenstrual Dysphoric Disorder" OR DE "Seasonal Affective Disorder" OR (DE "Major Depression" OR DE "Anaclitic Depression" OR DE "Dysthymic Disorder" OR DE "Endogenous Depression" OR DE "Late Life Depression" OR DE "Reactive Depression" OR DE "Recurrent Depression" OR DE "Treatment Resistant Depression" OR DE "Postpartum Depression") OR (depression or depressive disorder or depressive symptoms OR "affective disorder*" OR "depressive Disorder*" OR "depressive episode*" OR "depressive neuroses" OR "depressive neurosis" OR "depressive symptom*" OR "depressive syndrome" OR "depressive syndromes" OR "depression" OR "Emotional Depression" OR "Melancholia" OR "Melancholias" OR "mood disorder*" OR "dysthymic disorder*" OR "dysthymia*") Note: Affective Disorders/explode = DE "Affective Disorders" OR DE "Disruptive Mood Dysregulation Disorder" OR DE "Major Depression" OR DE "Persistent Depressive Disorder" OR DE "Premenstrual Dysphoric Disorder" OR DE "Seasonal Affective Disorder" Note: Major Depression/explode = (DE "Major Depression" OR DE "Anaclitic Depression" OR DE "Dysthymic Disorder" OR DE "Endogenous Depression" OR DE "Late Life Depression" OR DE "Postpartum Depression" OR DE "Reactive Depression" OR DE "Recurrent Depression" OR DE "Treatment Resistant Depression")
Executive Function	"Attention"[MeSH] OR "Cognition"[MeSH] OR "Memory"[MeSH] OR "Executive Function"[MeSH] OR "Problem Solving"[MeSH] OR "Decision Making"[Mesh] OR "attention"[All Fields] OR "cognition"[All Fields] OR "cognitions"[All Fields] OR "cognitive development" OR "cognitive flexibility" OR "cognitive function*" OR "decision making" OR "decision making"[Text Word] OR "decisionmaking"[Text Word] OR "executive function*" [Text Word] OR "memories"[Text Word] OR "memory"[Text Word] OR "problem solving"[Text Word] OR "problem solving"[TIAB:~2] OR "problemsolving"[Text Word] OR "executive function"[TIAB:~2]	(MM "Executive Function") OR executive function OR executive dysfunction OR executive functioning OR TI (attention OR cognition OR cognitions OR cognitive development OR cognitive flexibility OR cognitive function OR decision making OR executive Function OR executive functions OR memory OR memories OR problem Solving) OR AB (attention OR cognition OR cognitions OR cognitive development OR cognitive flexibility OR cognitive function OR decision making OR executive Function OR executive functions OR memory OR memories OR problem Solving) OR (SU attention OR cognition OR cognitions OR cognitive development OR cognitive flexibility OR cognitive function OR decision making OR executive Function OR executive functions OR memory OR memories OR problem Solving)	((MM "Executive Function") OR (MM "Cognitive Ability") OR (MM "Cognitive Flexibility")) OR (DE "Cognitive Impairment" OR DE "attention" OR DE "cognition" OR DE "cognitions" OR DE "cognitive development" OR DE "cognitive flexibility" OR DE "cognitive function" OR DE "decision making" OR DE "executive Function" OR DE "executive functions" OR DE "memory" OR DE "memories" OR DE "problem solving") OR (attention OR cognition OR cognitions OR cognitive development OR cognitive flexibility OR cognitive function OR decision making OR executive Function OR executive functions OR memory OR memories OR problem Solving)
Type 1 Diabetes	(Insulin Dependent Diabetes Mellitus OR Juvenile-Onset Diabetes Mellitus OR IDDM OR Juvenile-Onset Diabetes OR Juvenile Onset Diabetes OR Type 1 Diabetes Mellitus OR Insulin Dependent Diabetes Mellitus 1 OR Type 1 Diabetes OR Autoimmune Diabetes) OR ("Diabetes Mellitus, Type 1"[Mesh]) OR Insulin Dependent Diabetes Mellitus[Text Word] OR Juvenile-Onset Diabetes Mellitus[Text Word] OR IDDM[Text Word] OR Juvenile-Onset Diabetes[Text Word] OR Juvenile Onset Diabetes[Text Word] OR Type 1 Diabetes Mellitus[Text Word] OR Insulin Dependent Diabetes Mellitus 1[Text Word] OR Type 1 Diabetes[Text Word] OR Autoimmune Diabetes[Text Word]	(MM "Diabetes Mellitus, Type 1") OR type 1 diabetes or t1d or diabetes mellitus or juvenile diabetes or insulin-dependent diabetes OR TI (OR type 1 diabetes or t1d or diabetes mellitus or juvenile diabetes or insulin-dependent diabetes) OR AB (OR type 1 diabetes or t1d or diabetes mellitus or juvenile diabetes or insulin-dependent diabetes)	MM "Diabetes Mellitus" OR type 1 diabetes or t1d or diabetes mellitus or juvenile diabetes or insulin-dependent diabetes OR Insulin Dependent Diabetes Mellitus

agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The authors have no competing interests to report

References

[1] International Diabetes Federation. *Type 1 diabetes estimates in children and adults*. 2022; Available from: <https://diabetesatlas.org/atlas/t1d-index-2022/>.

- [2] E.A. Beverly, F. Osowik, Clinically significant depressive symptoms and high diabetes distress in adults with type 1 and type 2 diabetes in Appalachian Ohio, *J. Osteopath. Med.* 121 (10) (2021) 813–824.
- [3] V.S. Helgeson, et al., 14-Year longitudinal trajectories of depressive symptoms among youth with and without type 1 diabetes, *J. Psychiatr. Psychol.* 47 (10) (2022) 1135–1144.
- [4] S. Liu, et al., Association and familial coaggregation of childhood-onset type 1 diabetes With depression, anxiety, and stress-related disorders: a population-based cohort Study, *Diabetes Care* 45 (9) (2022) 1987–1993.
- [5] M.M. McCarthy, et al., Diabetes distress, depressive symptoms, and cardiovascular health in adults with type 1 diabetes, *Nurs. Res.* 68 (6) (2019) 445–452.
- [6] E. Silina, Prevalence of anxiety and depression symptoms in adolescents with type 1 diabetes (T1D) and their parents, *Nord. J. Psychiatry* (2022) 1.
- [7] C.J. Garey, et al., The association between depression symptom endorsement and glycemic outcomes in adolescents with type 1 diabetes, *Pediatr. Diabetes* 23 (2) (2022) 248–257.
- [8] R.J. Anderson, et al., The prevalence of comorbid depression in adults with diabetes: a meta-analysis, *Diabetes Care* 24 (6) (2001) 1069–1078.
- [9] P. Gilsanz, et al., Depression in type 1 diabetes and risk of dementia, *Aging Ment. Health* 23 (7) (2019) 880–886.
- [10] B. Johnson, et al., Prevalence of depression among young people with Type 1 diabetes: a systematic review, *Diabet. Med.* 30 (2) (2013) 199–208.
- [11] M. de Wit, F. Pouwer, F.J. Snoek, How to identify clinically significant diabetes distress using the problem areas in diabetes (PAID) scale in adults with diabetes treated in primary or secondary care? Evidence for new cut points based on latent class analyses, *BMJ Open* 12 (3) (2022) e056304.
- [12] K.R. Harrington, et al., Associations of diabetes self-management characteristics, HbA1c, and psychosocial outcomes with depressive symptoms in a contemporary sample of adolescents with type 1 diabetes, *J. Diabetes Complicat.* 35 (3) (2021) 107838.
- [13] K.M.C. Hong, et al., Glycemic control, depression, diabetes distress among adolescents with type 1 diabetes: effects of sex, race, insurance, and obesity, *Acta Diabetol.* 58 (12) (2021) 1627–1635.
- [14] B. Jurgens, et al., Associations between depressive symptoms, fear of hypoglycemia, adherence to management behaviors and metabolic control in children and adolescents with type 1 diabetes, *J. Clin. Psychol. Med. Settings* 27 (2) (2020) 385–395.
- [15] A. Schmitt, et al., Associations of depression and diabetes distress with self-management behavior and glycemic control, *Health Psychol. J.* 40 (2) (2021) 113–124.
- [16] M.K. Cho, M.Y. Kim, Associated factors with depression and sleep quality in T1DM patients: a cross-sectional descriptive study, *BMC Psychiatry* 23 (1) (2023) 18.
- [17] E.L. Tracy, et al., Relationship satisfaction moderates links between poor sleep quality and psychological distress among couples coping with type 1 diabetes, *Sleep. Health* 8 (6) (2022) 705–713.
- [18] A. Diamond, Executive functions, *Annu. Rev. Psychol.* 64 (2013) 135–168.
- [19] L. Zhang, et al., Association between Diabetes and Cognitive Function among People over 45 Years Old in China: A Cross-Sectional Study, *Int. J. Environ. Res. Public Health* 16 (7) (2019).
- [20] M.M. Broadley, M.J. White, B. Andrew, A systematic review and meta-analysis of executive function performance in type 1 diabetes mellitus, *Psychosom. Med.* 79 (6) (2017) 684–696.
- [21] P.A. Gaudieri, et al., Cognitive function in children with type 1 diabetes: A meta-analysis, *Diabetes Care* 31 (9) (2008) 1892–1897.
- [22] J. He, et al., Cognitive function and neuroimaging characteristics in patients with childhood-onset type 1 diabetes mellitus, *Diabetes/Metab. Res. Rev.* (2023) e3613.
- [23] T.T. Ly, et al., Neurocognitive outcomes in young adults with early-onset type 1 diabetes: A prospective follow-up study, *Diabetes Care* 34 (10) (2011) 2192–2197.
- [24] Q. Zhao, et al., Executive function and diabetes: A clinical neuropsychology perspective, *Front. Psychol.* 11 (2020) 2112.
- [25] K. Ding, et al., The relationship between executive functioning, type 1 diabetes self-management behaviors, and glycemic control in adolescents and young Adults, *Curr. Diabetes Rep.* 21 (3) (2021) 10.
- [26] K. McNally, et al., Executive functioning, treatment adherence, and glycemic control in children with type 1 diabetes, *Diabetes Care* 33 (6) (2010) 1159–1162.
- [27] W. Bai, et al., Inter-relationship between cognitive performance and depressive symptoms and their association with quality of life in older adults: A network analysis based on the 2017–2018 wave of Chinese Longitudinal Healthy Longevity Survey (CLHLS), *J. Affect. Disord.* 320 (2023) 621–627.
- [28] G.S. Brewster, et al., Depressive Symptoms, Cognition, and Everyday Function Among Community-Residing Older Adults, *J. Aging Health* 29 (3) (2017) 367–388.
- [29] Dehn, L.B. and T. Beblo, *[Depressed, biased, forgetful: The interaction of emotional and cognitive dysfunctions in depression]*. Neuropsychiatrie: Klinik, Diagnostik, Therapie und Rehabilitation: Organ der Gesellschaft Österreichischer Nervenärzte und Psychiater, 2019. 33(3): p. 123–130.
- [30] C. Dell'Acqua, et al., Depressive symptoms and cognitive control: The role of affective interference, *Cogn. Emot.* 36 (7) (2022) 1389–1403.
- [31] K.R. Noll, C.M. Sullaway, J.S. Wefel, Depressive symptoms and executive function in relation to survival in patients with glioblastoma, *J. Neuro-Oncol.* 142 (1) (2019) 183–191.
- [32] R.L. Morgan, et al., Identifying the PECO: A framework for formulating good questions to explore the association of environmental and other exposures with health outcomes, *Environ. Int.* 121 (Pt 1) (2018) 1027–1031.
- [33] M.J. Page, J.E. McKenzie, P.M. Bossuyt, I. Boutron, T.C. Hoffmann, C.D. Mulrow, L. Shamseer, J.M. Tetzlaff, E.A. Akl, S.E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M.M. Lalu, T. Li, E.W. Loder, E. Mayo-Wilson, S. McDonald, L.A. McGuinness, D. Moher, The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, *BMJ* (372) (2021) 71.
- [34] S. Bikri, et al., Predicting visual perception and working memory deficits among patients with type 1 diabetes: The implication of eating attitude and mental health status, *Acta Neuropsychol.* 19 (4) (2021) 501–519.
- [35] M.M. Broadley, M.J. White, B. Andrew, Executive function is associated with diabetes-specific disordered eating in young adults with type 1 diabetes, *J. Psychosom. Res.* 111 (2018) 1–12.
- [36] E. Duinkerken, et al., Subgenual cingulate cortex functional connectivity in relation to depressive symptoms and cognitive functioning in type 1 diabetes mellitus patients, *Psychosom. Med.* 78 (6) (2016) 740–749.
- [37] K.M. Perez, et al., Executive function in adolescents with type 1 diabetes: Relationship to adherence, glycemic control, and psychosocial outcomes, *J. Pediatr. Psychol.* 42 (6) (2017) 636–646.
- [38] M. Yin, et al., Effects of depression on cognitive function in patients with diabetes in different age groups, *J. Clin. Exp. Neuropsychol.* (2023).
- [39] T. Bell, et al., The association between cognitive decline and incident depressive symptoms in a sample of older Puerto Rican adults with diabetes, *Int. Psychogeriatr.* 29 (8) (2017) 1317–1325.
- [40] C.-J. Chiu, et al., The short- and long-term effects of psychobehavioral correlates in buffering diabetes-related cognitive decline, *Ann. Behav. Med.* 50 (3) (2016) 436–444.
- [41] B. Downer, et al., Effects of comorbid depression and diabetes mellitus on cognitive decline in older Mexican Americans, *J. Am. Geriatr. Soc.* 64 (1) (2016) 109–117.
- [42] A.M. Rotkiewicz-Piorun, et al., Cognitive decline in older Mexican Americans with diabetes, *J. Natl. Med. Assoc.* 98 (11) (2006) 1840–1847.
- [43] G.C. Ciuhău, D. Iliescu, Depression and learning problems in children: Executive function impairments and inattention as mediators, *Acta Psychol.* 220 (2021) 103420.
- [44] V.M. Dotson, et al., Depression and cognitive control across the lifespan: A systematic review and meta-analysis, *Neuropsychol. Rev.* 30 (4) (2020) 461–476.
- [45] P.L. Rock, et al., Cognitive impairment in depression: a systematic review and meta-analysis, *Psychol. Med.* 44 (10) (2014) 2029–2040.